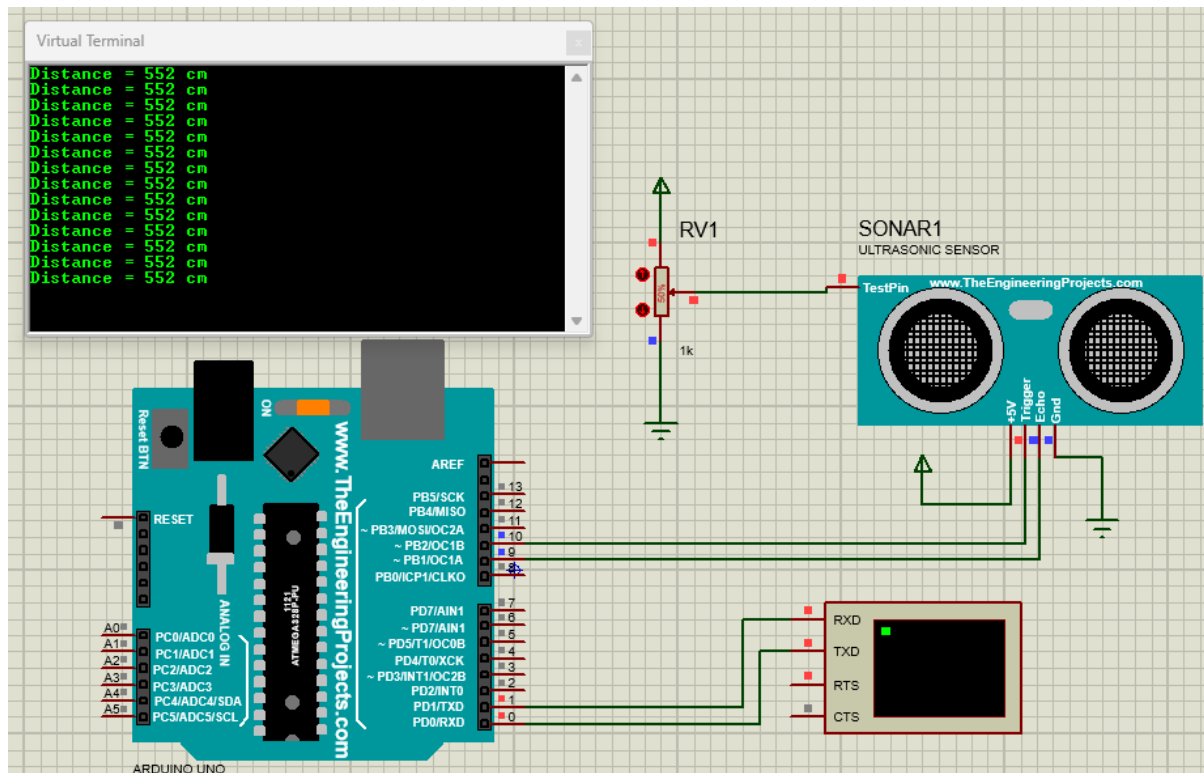


Exp No: 9: ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

CIRCUIT DIAGRAM:



PROGRAM:

// Ultrasonic Distance Measurement using HC-SR04

```
int trig = 10;
```

```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup() {
```

```
    pinMode(trig, OUTPUT);
```

```
    pinMode(echo, INPUT);
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
    // Send ultrasonic trigger pulse
```

```
    digitalWrite(trig, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(trig, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(trig, LOW);
```

```
    // Measure echo time
```

```
    duration = pulseIn(echo, HIGH);
```

```
    // Calculate distance
```

```
    cm = (duration / 2) * 0.034;
```

```
    // Display distance
```

```
    Serial.print("Distance = ");
```

```
    Serial.print(cm);
```

```
    Serial.println(" cm");
```

```
    delay(500);
```

```
}
```